

ALTAML Assignment

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Task 1: Convert the following pseudo Ruby code in your chosen language(Python).

A) cat.py

```
src > altaml > cat.py > Cat
1 class Cat:
2     def __init__(self):
3         self.name = None
4         self.age = None
5         self.favoriteFood = None
6
7
8     def getName(self):
9         return self.name
10
11
12     def getAge(self):
13         return self.age
14
15
16     def getFavoriteFood(self):
17         return self.favoriteFood
18
19     def setName(self, newName):
20         self.name = newName
21
22     def setAge(self, newAge):
23         self.age = newAge
24
25     def setFavoriteFood(self, newFavoriteFood):
26         self.favoriteFood = newFavoriteFood
27
```

B) data.py

```
src > altaml > data.py > Data > insert
1 class Data:
2     def __init__(self, database):
3         print("Connecting to database")
4
5     def beginTran(self):
6         print("Beginning a transaction")
7
8     def commit(self):
9         print("Committing transaction")
10
11     def rollback(self):
12         print("Rolling back transaction")
13
14     def insert(self, table, object):
15         print("Inserting {0} into table {1}".format(object.getName(), table))
```

C) main.py

```
src > altaml > main.py > ...
1  from .cat import Cat
2  from .data import Data
3
4
5  cat = Cat()
6  cat_name = cat.getName()
7
8  if cat_name is None:
9      print("Name is currently ")
10 else:
11     print("Name is currently {}".format(cat_name))
12
13
14 cat.setName("Garfield")
15 cat_name = cat.getName()
16 print("Name has been changed to {}".format(cat_name))
17
18 data = Data("database")
19 data.insert(["Cat", cat])
```

Expected outcome –

```
Problems  Output  Debug Console  Terminal  Ports
• (altaml) PS C:\Users\sharm\Desktop\AltaML> uv run python -m altaml.main
Name is currently
Name has been changed to Garfield
Connecting to database
Inserting Garfield into table Cat
○ (altaml) PS C:\Users\sharm\Desktop\AltaML> 
```

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Task 2:

A) Modify the constructor to set the cat's initial age to a random number between 5 and 10.

```
1 import random
```

```
2 def __init__(self, name):
```

```
7         self.age = random.randint(5,10)
```

B) Modify the constructor to support an optional initial name for the new cat.

```
5         def __init__(self, name = None):
```

```
6             self.name = name
```

C) add a method called speak() to make the cat say "meow" (use print or echo to speak).

```
49         def speak(self, sound = None):
```

D) Modify the new speak() method to accept an optional argument.

If not supplied, the cat still says "meow".

If supplied, the cat prints the value of the argument.

```
49         def speak(self, sound = None):
```

```
54             if sound is None :
```

```
55                 print("meow")
```

```
56             else:
```

```
57                 print(sound)
```

E) Modify the setName() method to keep a list of all the names the cat has ever had

```
5  def __init__(self, name = None):
6      self.name = name
7      self.age = random.randint(5,10)
8      self.favoriteFood = None
9      self.history_name=[]
10     self.speak_count = 0
11
12     if name is not None:
13         self.history_name.append(self.name)
```

```
38  def setName(self, newName):
39      if newName is not None:
40          self.history_name.append(newName)
41          self.name = newName
42
```

F) Create a getNames() method to return a list of all the names the cat has ever had

```
15  def getName(self):
16      return self.name
```

G) Modify the class so that the cat's age increases by 1 every five times it speaks.

```
✓ def __init__(self, name = None):  
    self.name = name  
    self.age = random.randint(5,10)  
    self.favoriteFood = None  
    self.history_name=[]  
    self.speak_count = 0
```

```
49 def speak(self,sound = None):  
50     self.speak_count=self.speak_count+1  
51     if self.speak_count%5==0:  
52         self.age+=1  
53  
54     if sound is None :  
55         print("meow")  
56     else:  
57         print(sound)  
58
```

H) Create a new method getAverageNameLength() that returns the average length of all the names the cat has ever had.

```
27 def getAverageNameLength(self):  
28     if len(self.history_name) == 0:  
29         return 0.0  
30  
31     total_len = 0  
32     for i in (self.history_name):  
33         current_name = i  
34         total_len += len(current_name)  
35  
36     return total_len/len(self.history_name)
```

l) Make a Dog object, making necessary dog-like changes.

```
src > altaml > dog.py > Dog > speak
1  import random
2
3  class Dog:
4
5      def __init__(self, name = None):
6          self.name = name
7          self.age = random.randint(5,10)
8          self.favoriteFood = None
9          self.history_name=[]
10         self.speak_count = 0
11
12         if name is not None:
13             self.history_name.append(self.name)
14
15     def getName(self):
16         return self.name
17
18     def getAge(self):
19         return self.age
20
21     def getFavoriteFood(self):
22         return self.favoriteFood
23
24     def getNames(self):
25         return list(self.history_name)
26
27     def getAverageNameLength(self):
28         if len(self.history_name) == 0:
29             return 0.0
30
31         total_len = 0
32         for i in (self.history_name):
33             current_name = i
34             total_len += len(current_name)
35
36         return total_len/len(self.history_name)
37
38     def setName(self, newName):
39         if newName is not None:
40             self.history_name.append(newName)
41         self.name = newName
42
43     def setAge(self, newAge):
44         self.age = newAge
45
46     def setFavoriteFood(self, newFavoriteFood):
47         self.favoriteFood = newFavoriteFood
48
49     def speak(self, sound = None):
50         self.speak_count=self.speak_count+1
51         if self.speak_count%5==0:
52             self.age+=1
53
54         if sound is None :
55             print("bark")
56         else:
57             print(sound)
58
```